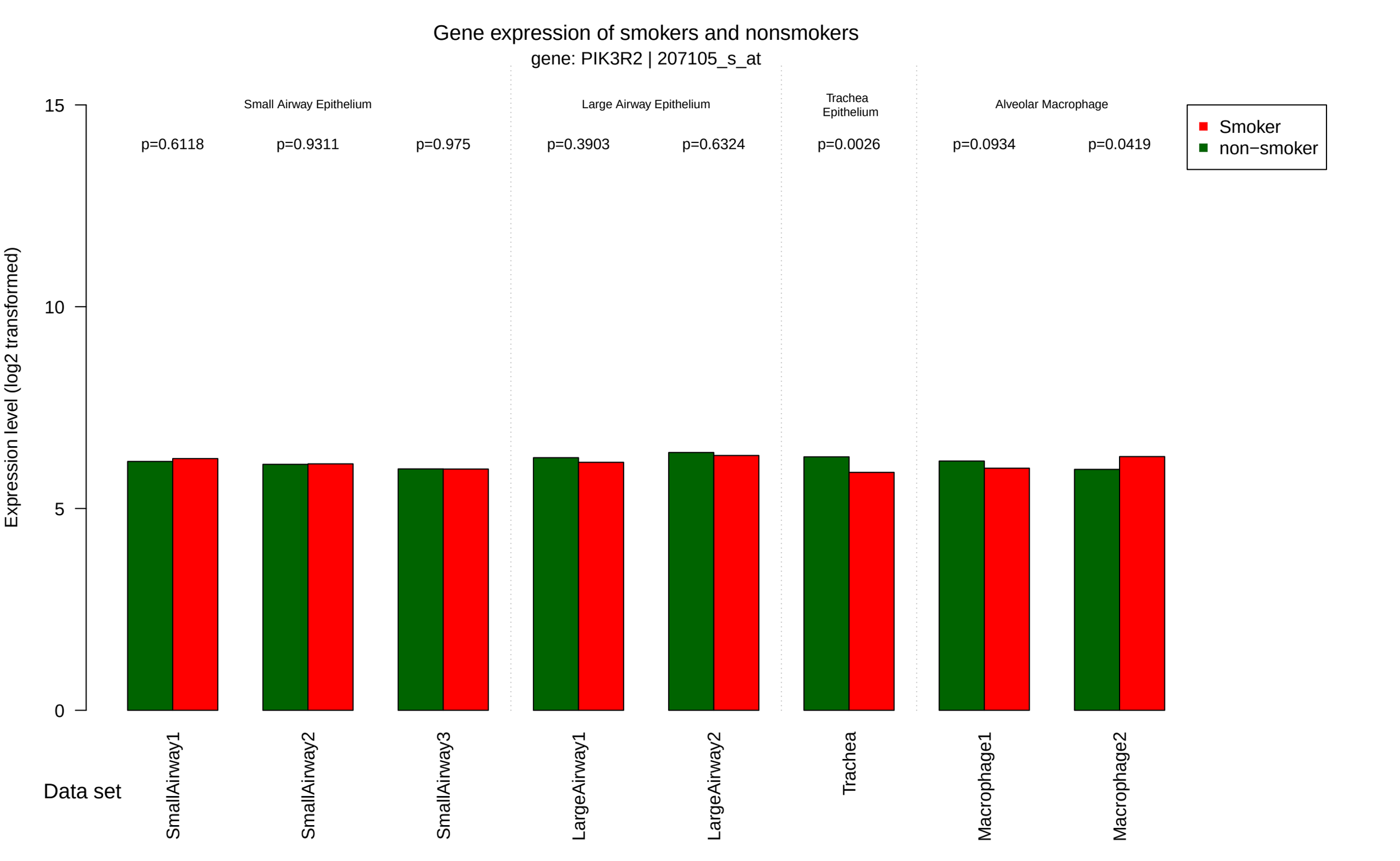
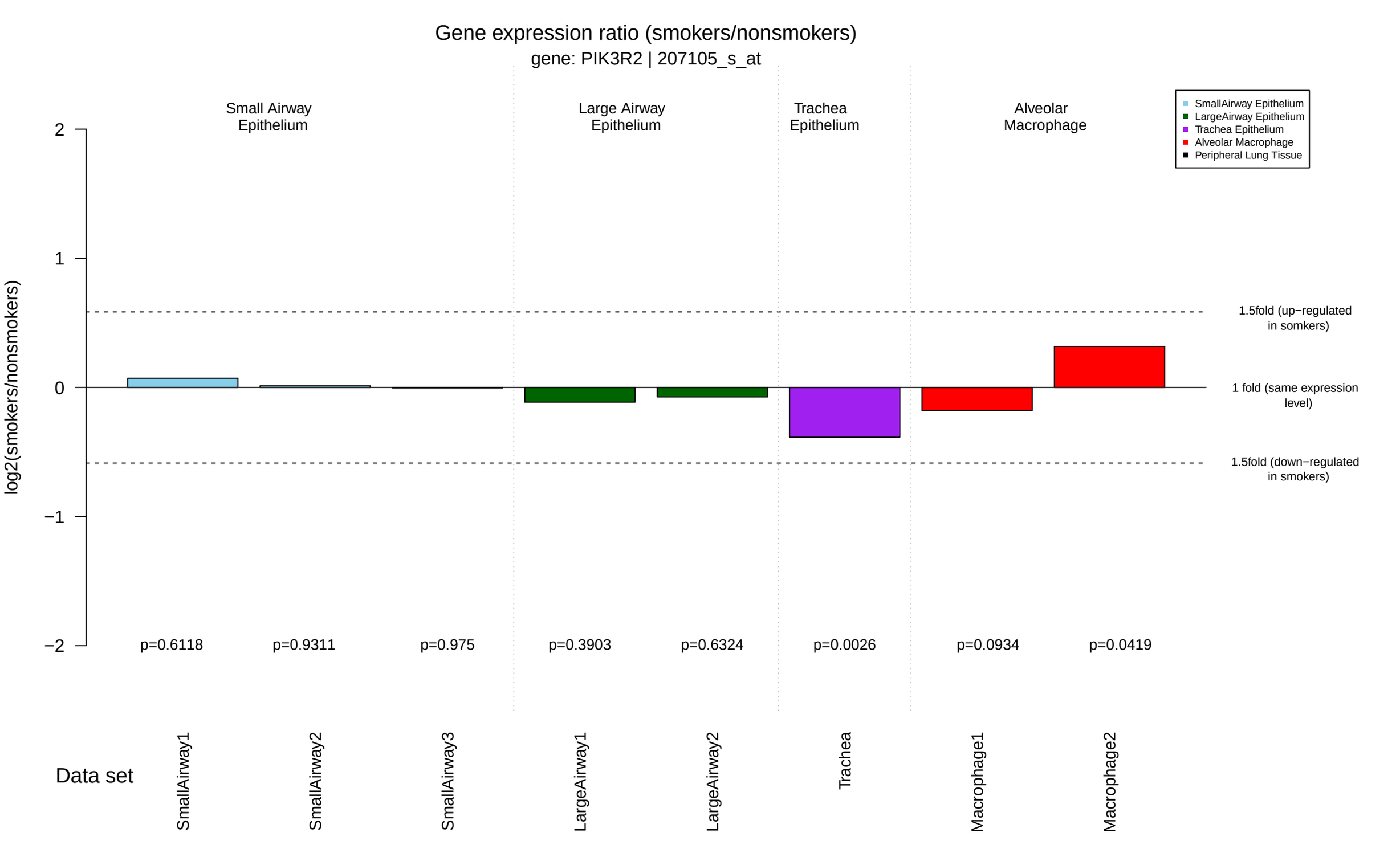
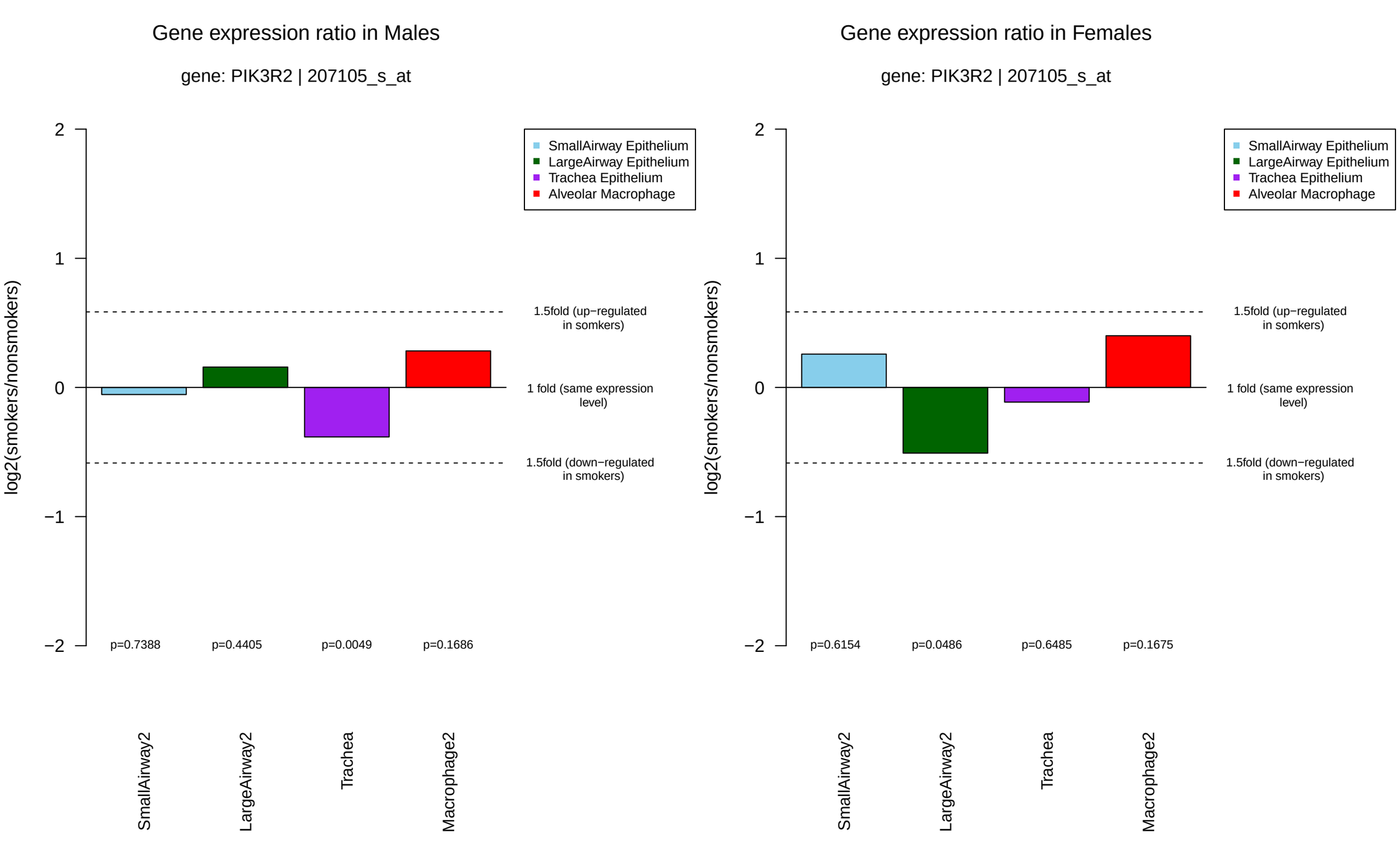
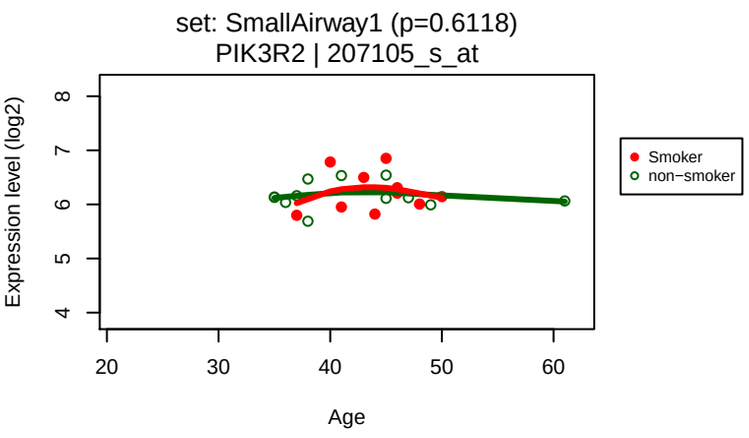
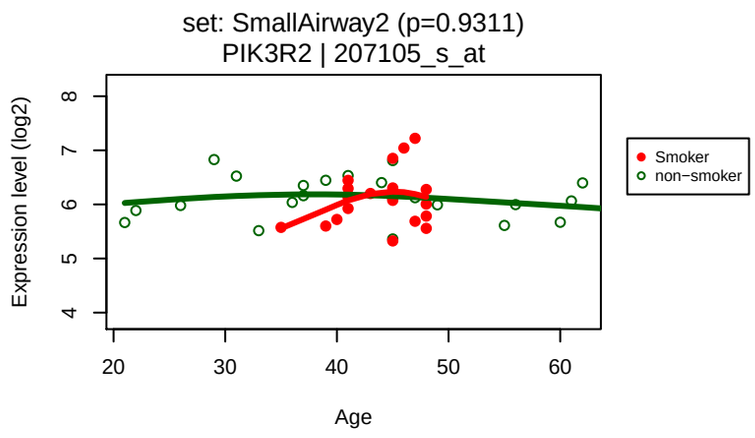
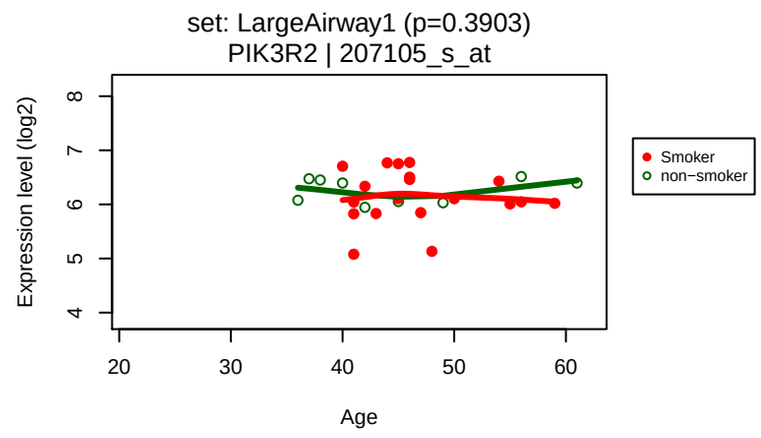

Gene expression of smokers and nonsmokers
gene: PIK3R2 | 207105_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6118
p=0.9311
p=0.975
p=0.3903
p=0.6324
p=0.0026
p=0.0934
p=0.0419
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

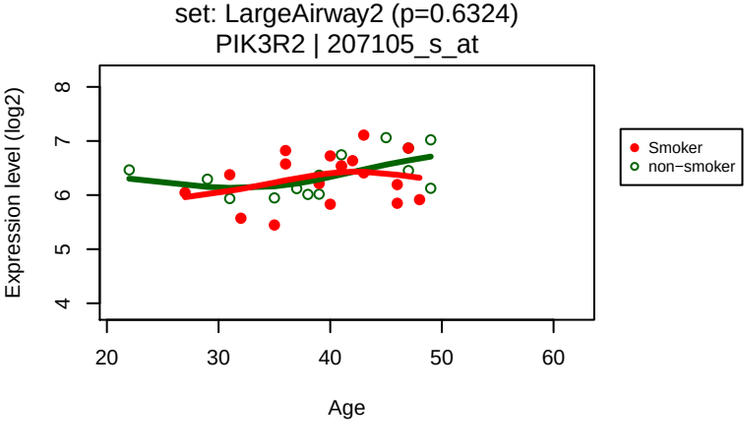

Gene expression ratio (smokers/nonsmokers)
gene: PIK3R2 | 207105_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6118
p=0.9311
p=0.975
p=0.3903
p=0.6324
p=0.0026
p=0.0934
p=0.0419
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

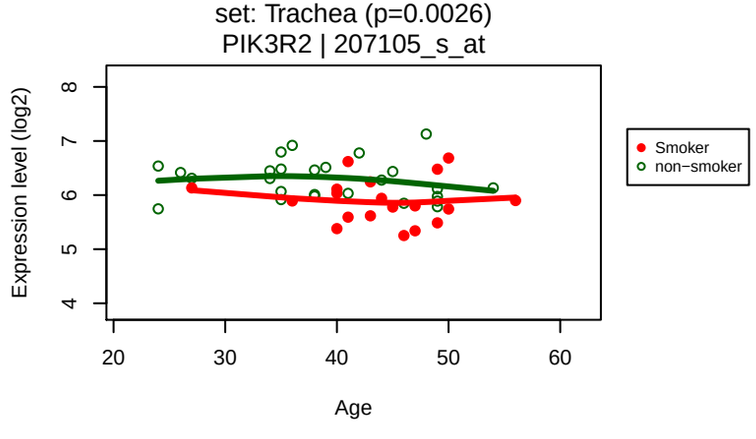

Gene expression ratio in Males
gene: PIK3R2 | 207105_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7388
p=0.4405
p=0.0049
p=0.1686
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3R2 | 207105_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6154
p=0.0486
p=0.6485
p=0.1675
SmallAirway2
LargeAirway2
Trachea
Macrophage2

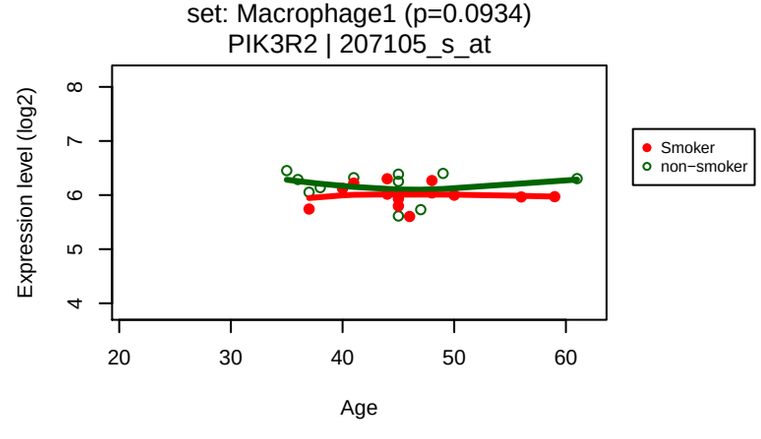

set: SmallAirway1 (p=0.6118)
PIK3R2 | 207105_s_at
Expression level (log2)
Age
Smoker
non−smoker

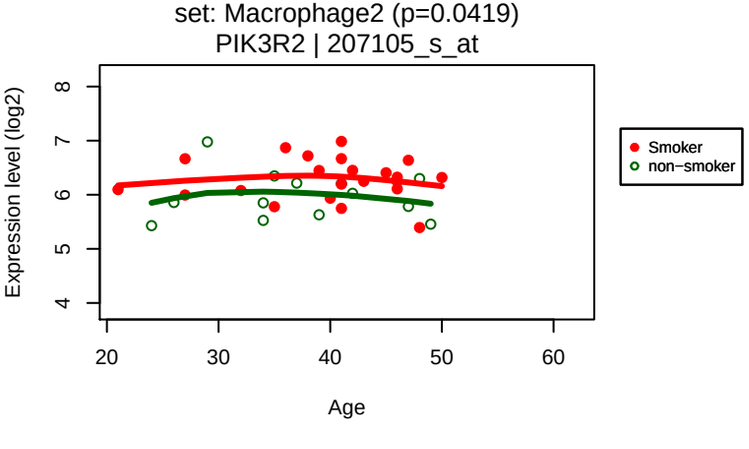

set: SmallAirway2 (p=0.9311)
PIK3R2 | 207105_s_at
Expression level (log2)
Age
Smoker
non−smoker

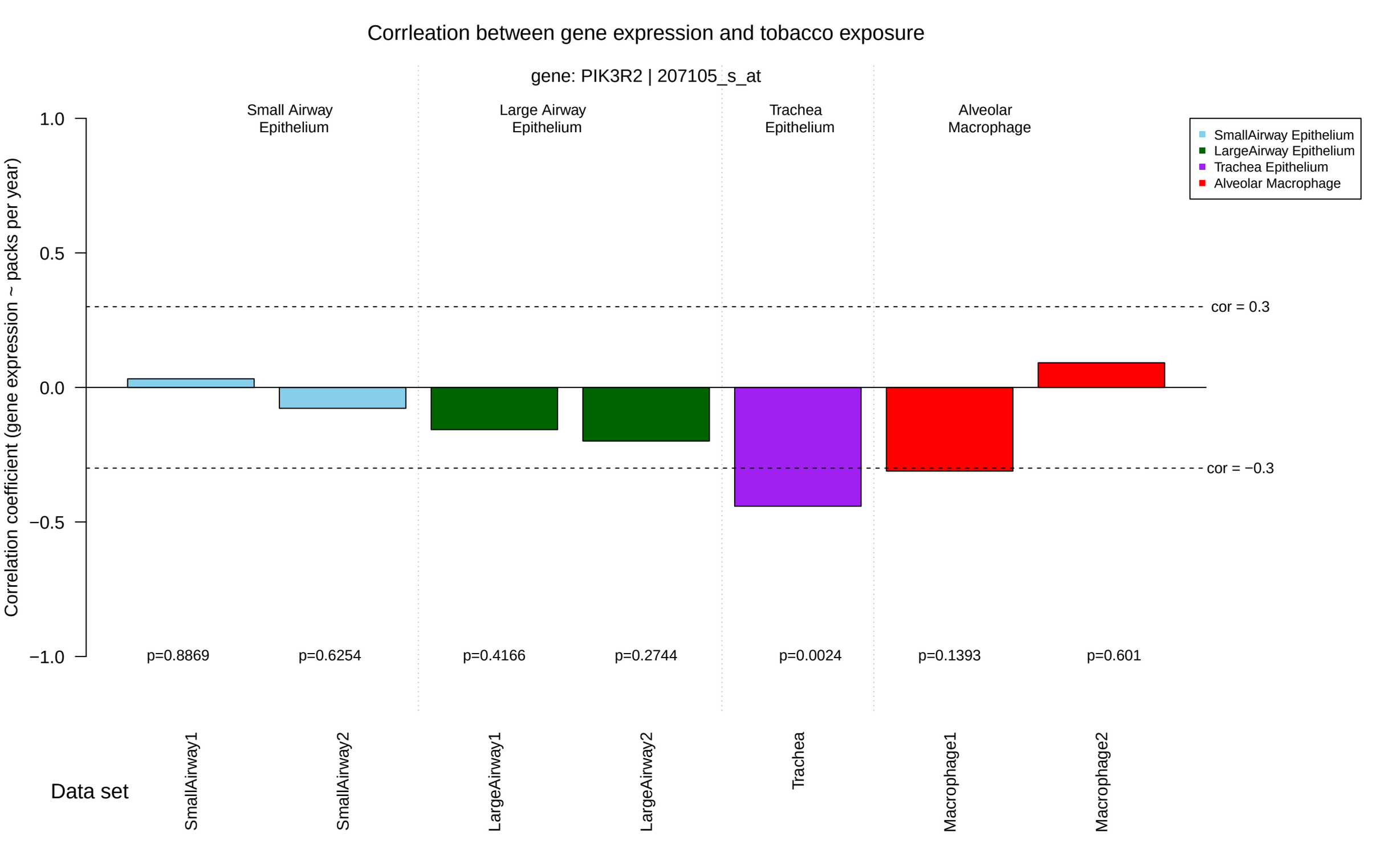

set: LargeAirway1 (p=0.3903)
PIK3R2 | 207105_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6324)
PIK3R2 | 207105_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0026)
PIK3R2 | 207105_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0934)
PIK3R2 | 207105_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0419)
PIK3R2 | 207105_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3R2 | 207105_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8869
p=0.6254
p=0.4166
p=0.2744
p=0.0024
p=0.1393
p=0.601
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2